

Consider that the frequency of rare events cannot be estimated from empirical observation for the very reason that *they are rare*. We thus need a prior model representation for that; the rarer the event, the higher the error in estimation from standard inductive methods (say, frequency sampling from counting past occurrences), hence the higher the dependence on an a priori representation that extrapolates into the space of low-probability events (which necessarily are not seen often).^{*}

But even outside of small probabilities, the a priori problem is always present. It seems salient with respect to rare events, but it pervades probabilistic knowledge. I will present two versions I have been working on with two collaborators, Avital Pilpel, a philosopher of science (he walks fast), and Raphael Douady, a mathematician (he is sometimes a good walker, when he is not busy).

Epimenides the Cretan

Avital Pilpel and I expressed the regress argument as follows, as the epistemic problem of risk management, but the argument can be generalized to any form of probabilistic knowledge. It is a problem of *self-reference* by probability measures.

We can state it in the following way. If we need data to obtain a probability distribution to gauge knowledge about the future behavior of the distribution from its past results, and if, at the same time, we need a probability distribution to gauge data sufficiency and whether or not it is predictive of the future, then we face a severe regress loop. This is a problem of self-reference akin to that of Epimenides the Cretan stating whether or not Cretans are liars. Indeed, it is too uncomfortably close to the Epimenides situation, since a probability distribution is used to assess the degree of truth but cannot reflect on its own degree of truth and validity. And, unlike many problems of self-reference, those related to risk assessment have severe consequences. The problem is more acute with small probabilities.

An Undecidability Theorem

This problem of self-reference, published with Pilpel after *The Black Swan*, went unnoticed as such. So Raphael Douady and I re-expressed the philosophical problem mathematically, and it appears vastly more devastating in its practical implications than the Gödel problem.

Raphael is, among the people I know, perhaps the man with the